

Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Does Club Wage Spending Predict Premier League
Finishing Position? (2025-26)

Full Name: Olzhabay Kuanysh

ID: 24B031954

06.10.2026

Almaty

Does Club Wage Spending Predict Premier League Finishing Position? (2025-26)

SIS №1 report – data collection, preparation and analysis

1. Goal

I test whether clubs that spend more on player wages finish higher in the Premier League. The analysis covers all 20 clubs of the 2025-26 season. The code is in the notebook SIS_OK_Wages_vs_Position.ipynb.

2. Data and approach

- **API:** football-data.org, endpoint `/v4/competitions/PL/standings` (season 2025). It gives the final table: position, points, goal difference.
- **Web scraping:** FBref “Squad Wages” table (payroll data from Capology), read with *requests* and *BeautifulSoup* after checking robots.txt. It gives the weekly and annual wage bill of each club in GBP. FBref’s bot protection returned HTTP 403 for the live request, so the notebook parsed a copy of the same page saved from the browser.
- **Cleaning:** extracted the GBP figure from text such as “£ 4,305,300 (€ ...)”, fixed data types, renamed columns, and checked missing values and duplicates (none found). Club names are spelled differently on the two sites, so I mapped them to the API’s 3-letter club codes.
- **Merging:** one-to-one join on the club code, giving 20 clubs. **Analysis:** descriptive statistics, Pearson correlation and linear regression (wages vs points), Spearman rank correlation (wage rank vs final position), and the gap “wage rank minus final position” to find over- and under-performers.

3. Results

Measure	Value	Reading
Wage bill vs points (Pearson r)	0.68 ($p < 0.01$)	Strong positive
Variation in points explained (R^2)	0.47	About half
Regression slope	+0.21 points per £1m	About +2 points per £10m
Wage bill vs goal difference (Pearson r)	0.73	Strong positive
Wage rank vs final position (Spearman ρ)	0.46 ($p = 0.04$)	Moderate positive

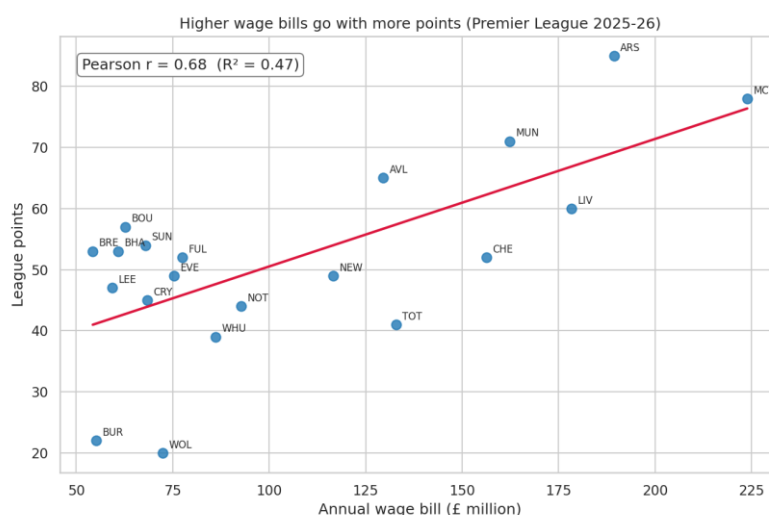


Figure 1. Annual wage bill vs league points; the red line is the linear fit.

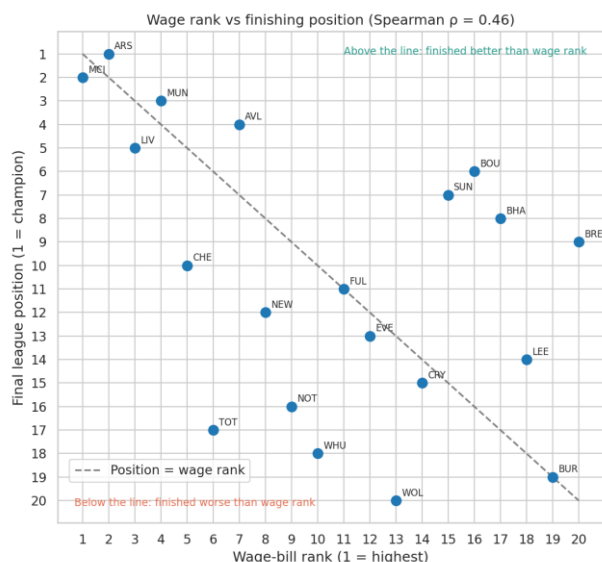


Figure 2. Wage rank vs final position (dashed line = perfect match).

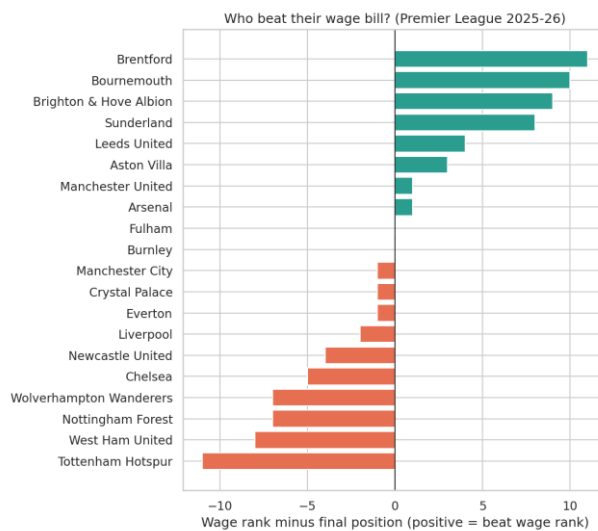


Figure 3. Places gained (+) or lost (–) relative to wage rank.

4. Key insights

- **Wages predict points well, positions only moderately.** The top 5 finishers all had a top-7 wage bill, and the six biggest wage bills averaged 64.5 points against 46.4 for the other 14 clubs. In the tightly packed middle of the table, small point gaps reshuffle positions, which lowers the rank correlation.
- **Over-performers:** Brentford (20th-highest wage bill, finished 9th), Bournemouth (16th to 6th), Brighton (17th to 8th) and Sunderland (15th to 7th). Brentford bought points cheapest, about £1.0m per point.
- **Under-performers:** Tottenham (6th to 17th), West Ham (10th to 18th), Nottingham Forest and Wolves (7 places worse each) and Chelsea (5th to 10th). Wolves paid the most per point, about £3.6m.
- **Answer:** wage spending is a useful but incomplete predictor. Money gives the big spenders a high floor, but it does not guarantee a place.

5. Limitations

Only one season and 20 clubs, so the results are indicative. Capology wages are partly estimates (44–100% of each club's figure). Wage bills exclude transfer fees, and correlation does not prove causation: richer clubs can pay more and also buy better players.